Habib University

Introduction to Probability and Statistics – EE 354/CE 361/MATH 310 – Fall 2025

Instructor – Dr. Aamir Hasan - Section L2

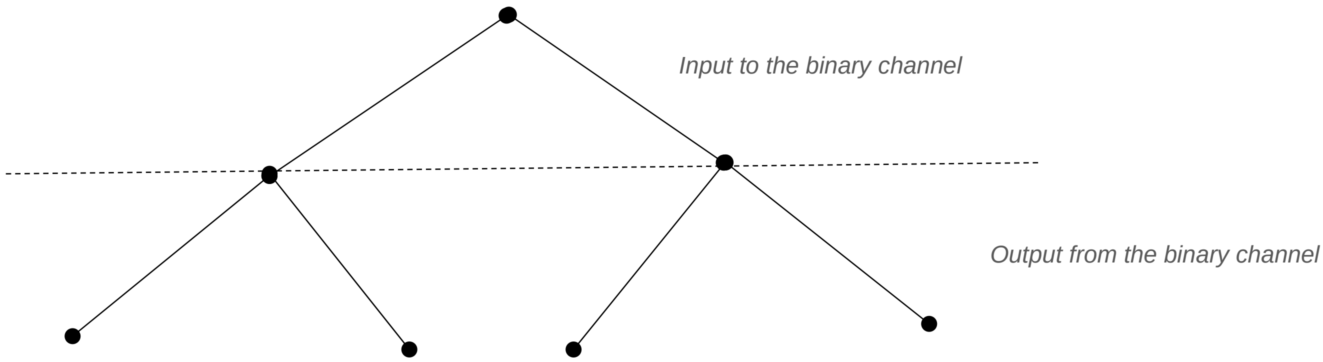
Time = 20 Minutes **Quiz No 2** [20 points]

Note - *Provide all your answers on the answer sheet. Present your final answers and reasoning clearly. Most of the total score for the questions will be awarded on your reasoning. Please refrain from random or incoherent scribbles as they will not be graded.* ***Question No 2 is on page 2***.

Question 1 [10 points] Conditional Probability

Many communication systems can be modelled in the following way. First, the user inputs a 0 or a 1 into the system, and a corresponding signal is transmitted. Second, the receiver makes a decision about what was the input to the system, based on the signal strength it received. Suppose that the user sends 0s with probability $(1-p)$ and 1s with probability p, and suppose that the receiver makes random decision errors with probability ε (incorrect decision by the receiver). For $i = 0, 1,$ let A_i be the event "input was i," and let B_i be the event "receiver decision was i."

By completing the following tree diagram, i.e., ***mark the probabilities of input-output pairs in the binary transmission system***, find the probabilities for $P(A_i \cap B_j)$ for $i = 0, 1,$ and $j = 0, 1$. (***Note – you need to find 4 probabilities***).



Question 2 [10 points] Bayes Rule

In the binary communication system, in Question 1, we will try to find out which input is more probable given that the receiver has output a 1. Assume that, a priori, the input is equally likely to be 0 or 1 (i.e., $p = 0.5$). Let A_k be the event that the input was k, for $k = 0, 1$. ***Let B_1 be the event "receiver output was a 1"***. Assume probability of receiver makes random decision errors with probability ε - similar to question No 1.

(a) Find the probability of event B_1, P$[B_1]$. Hint - you may use the tree diagram of question No 1 and using the 'Total Probability Theorem'.

(b) Using Bayes rule, find P$[A_0|B_1]$ and P$[A_1|B_1]$.

(c) Now assuming $\varepsilon = 0.1$, find which input is more probable given that the receiver has output a 1. How would your answer change if $\varepsilon = 0.5$.